

SMITKUMAR JAYENDRAKUMAR VELANI

📍 Boston, MA | 📞 +1 857-426-7025 | ✉ velanismitkumar@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 🌐 Portfolio

Education

Northeastern University, Boston, MA

Master of Science in Data Science

2026 – 2028

GPA: 4.0/4.0

Adani Institute of Infrastructure Engineering (GTU)

B.E. Information Communication & Technology

2021 – 2025

CGPA: 8.24/10

Work Experience

Unified Mentor Pvt. Ltd. | Data Analyst Intern Certificate

Jan 2025 – Apr 2025

Stock Market Data Analysis System 🔗 Link

Python | Pandas | NumPy | Matplotlib | Seaborn | yfinance | Tkinter

- Integrated **Yahoo Finance API** to fetch real-time **OHLCV** data for **4+ stocks** across **250+ daily records**.
- Engineered **11 technical indicators** (SMA, EMA, RSI, MACD, Bollinger Bands) using vectorised **NumPy**, no TA library.
- Built **Tkinter dashboard** with **20 clickable charts** computing **Sharpe Ratio** and **Max Drawdown** metrics.
- Automated data cleaning, deduplication and validation pipeline using **Pandas** across **4+ stock feeds**.

IBM SkillsBuild | Data Analytics Intern (CSRBOX) Certificate

Jun 2024 – Aug 2024

Plant Health Prediction System 🔗 Link

Python | NumPy | Flask | TensorFlow | OpenCV | Keras

- Deployed **MobileNetV2** CNN achieving **89% diagnostic accuracy** across **15 disease classes** on **PlantVillage**.
- Built **OpenCV** pipeline (HSV lesion analysis, Canny edge detection) with **two-phase training** and **EarlyStopping**.
- Applied **data augmentation** and **ReduceLROnPlateau** to improve CNN generalisation across **3 plant species**.
- Deployed end-to-end **Flask** app with drag-and-drop interface for real-time CNN inference.

Projects

Global Economic Intelligence Dashboard 🔗 Link

Full-Stack AI | Data Science Project

Python | Flask | D3.js | Scikit-Learn | Groq LLaMA | ARIMA

- Integrated **World Bank API** (190+ countries) with **ARIMA** forecasting, **Isolation Forest** and **VADER** sentiment.
- Deployed **D3.js** choropleth map with metric toggle, **K-Means** clustering and animated **GDP race** (2000-2024).
- Built **recession risk predictor** scoring 0-100% from GDP growth, inflation and unemployment via **Scikit-Learn**.
- Integrated **Groq LLaMA 3.3** for AI economic summaries and risk ratings with **WebSocket** via **Flask-SocketIO**.

RAG PDF Chatbot 🔗 Link

NLP | LLM | RAG Project

Python | LangChain | FAISS | HuggingFace | Gemini API

- Built **RAG pipeline** via **LangChain** processing PDFs into **500-word overlapping chunks** stored in **FAISS**.
- Implemented **HuggingFace** (all-MiniLM-L6-v2) semantic search retrieving **top-3 chunks** per query via cosine similarity.
- Integrated **Gemini 2.5 Flash** with **prompt engineering** to generate context-grounded answers from retrieved docs.
- Achieved factual responses on **15-page** research PDFs using **384-dim vectors** and **50-word** chunk overlap.

Loan Default Prediction with MLflow 🔗 Link

MLOps | ML Pipeline Project

Python | XGBoost | MLflow | Scikit-Learn | Pandas

- Trained 4 ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost) on **32,581 records** via **MLflow**.
- **XGBoost** outperformed all models achieving **93.4% accuracy** and **AUC: 0.947** on **Kaggle Credit Risk Dataset**.
- Performed end-to-end preprocessing (imputation, outlier removal, label encoding, scaling) on real loan data.
- Identified **loan grade**, **home ownership** and **loan percent income** as top predictors; saved via **joblib**.

Multi-Agent Atari Boxing with Deep Q-Learning 🔗 Link

Reinforcement Learning Project

Python | PyTorch | PettingZoo | OpenCV | Matplotlib

- Trained two **DQN agents** via **PettingZoo** and **PyTorch** over **200 episodes** on **T4 GPU** achieving **62%/38%** split.
- Built **CNN-based DQN** (3 Conv + 2 FC) on **4 stacked 84×84 frames**; implemented **Experience Replay**, **Target Network**.
- Applied **Epsilon-Greedy** (ϵ : 1.0 $\rightarrow$ 0.1) decay and preprocessed **RGB frames** via **OpenCV** for stable convergence.

DataPulse Analytics Dashboard 🔗 Link

Data Visualization Project

Python | Pandas | D3.js | JavaScript | HTML | CSS

- Built **7 interactive chart types** from scratch using **D3.js v7** SVG rendering across **11,900+ records**.
- Visualized **3 real-world datasets** (e-commerce, climate, stock market) with zero backend and client-side loading.

Technical Skills

Languages: Python, SQL, JavaScript, PHP, HTML, CSS, SwiftUI

Libraries: Pandas, NumPy, Matplotlib, Scikit-Learn, TensorFlow, Keras, PyTorch, OpenCV, Flask, LangChain, D3.js

Tools & Databases: PostgreSQL, MySQL, MongoDB, FAISS, MLflow, Power BI, Git, GitHub, VS Code

Machine Learning: CNN, Transfer Learning, DQN, RAG, LLM Integration, K-Means, ARIMA, Anomaly Detection, Semantic Search